

Resource leveling and float

In **resource leveling**, we look at **Total Float**, not **Free Float**.

Why?

- **Resource leveling** focuses on resolving **resource constraints** (limited manpower, machines, etc.).
- To delay activities **without delaying the overall project completion**, we must use the flexibility allowed by **Total Float**.
- **Total Float** tells us **how much an activity can be delayed without affecting the project finish date**.

Quick comparison (easy to remember):

Float Type	Used in	Reason
Total Float	Resource Leveling	Maintains project finish date
Free Float	Resource smoothing	Does not delay successor activities

In resource leveling, Total Float is considered because activities are shifted within their total float to resolve resource conflicts without delaying the project completion date.

Key correction

In true Resource Leveling, project duration *can* increase.

Therefore, **Total Float** may be exhausted or even become zero/negative.

So what float do we see in Resource Leveling?

We primarily consider **TOTAL FLOAT**, but

Resource leveling is **NOT** restricted by float availability

Correct conceptual distinction

Resource Smoothing

- Resources are adjusted **within available float**
- **Project duration does NOT change**
- Uses **Total Float**
- Finish date remains fixed

Resource Leveling

- Resources are limited
 - Activities may be delayed **beyond Total Float**
 - **Project duration CAN increase**
 - Critical path **may change**
 - Float concept becomes **secondary**
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Exam-safe table (very important)

Aspect	Resource Smoothing	Resource Leveling
Resource constraint	No	Yes
Project duration	Fixed	May increase
Float usage	Total Float only	May exceed float
Critical path	Same	Can change
Finish date	Not delayed	Can be delayed

In resource leveling, Total Float is considered initially, but activities may be delayed beyond float due to resource constraints, resulting in an increased project duration.

"Resource leveling is always performed within total float."

FALSE

Short, direct answer

Free Float is NOT used in resource leveling because it only protects successor activities, not the project completion date.

Clear conceptual reason (step-by-step)

♦ What Free Float really means

Free Float = delay allowed without delaying the *immediate successor*

It says **nothing** about:

- overall project finish date
- resource availability
- downstream resource conflicts

So it is **too local**.

♦ What Resource Leveling is trying to protect

Resource leveling tries to:

- resolve **resource over-allocation**
- manage **project-wide resource feasibility**
- accept that **project duration may change**

This requires **global flexibility**, not local flexibility.

Total Float measures global schedule flexibility, hence it is relevant.

Why Free Float fails in Resource Leveling

Issue	Free Float	Total Float
Scope	Local (between two activities)	Global (whole project)
Protects successors	Yes	Yes
Protects project finish	No	Yes
Handles resource conflicts	No	Yes
Used in leveling		

Critical point

Even when:

- an activity has **Free Float = 0**
- it may still have **Total Float > 0**

Resource leveling may delay it because:

- delaying it **does not immediately break the project**
- resources are more important than local precedence

One-line answer for theory questions

Free float is not used in resource leveling because it only ensures successor activities are not delayed, whereas resource leveling requires project-wide flexibility, which is measured by total float.

"Free float is preferred in resource leveling because it avoids delaying successors."

FALSE

Core truth

Resource leveling is NOT based on float at all.

Activities can be delayed even if Free Float = 0 and Total Float = 0.

Project finish date CAN change.

Then the real question:

Why still NOT Free Float?

Because Free Float has NO meaning once dates are allowed to change.

Key reasoning

♦ Free Float definition assumes:

- fixed early start of successor
- fixed network logic
- fixed project dates

! Resource leveling violates all three:

- successors may be delayed
- critical path may change
- finish date may move

Therefore:

Free Float becomes irrelevant / invalid in resource leveling

Why Total Float is *still mentioned* in books

- Total Float shows **initial schedule flexibility**
- It helps identify **less critical activities to delay first**
- But it is **NOT a constraint**

Once resources dominate:

- float is ignored
- resource feasibility dominates

In resource leveling, activities may be delayed beyond both free and total float due to resource constraints. Free float is not considered because it assumes fixed successor dates, which do not apply when project duration is allowed to change.

One-line memory hook

Free Float protects successors; Resource Leveling sacrifices successors and dates for resources.

Examiner trap clarified

Statement	Verdict
Resource leveling uses free float	False
Resource leveling always respects total float	False
Resource leveling may ignore float completely	True
